

Komponente, SMD	Kods "TME"
C 680 uF	EEVFK1J681M
C 220 uF	EEEFK1H221P
Inductor 33uH	DR74-330-R
LM2596	LM2596R-ADJ-TT

Komponente, SMD	Kods "UGE electronics"
Diode 1N5824	D17-1-1N5824-SMD

Plāte

The screenshot shows the 'Calculator Tools' application with the 'Track Width' tab selected. The 'Parameters' section on the left contains the following input values:

- Current (I): 3 A
- Temperature rise (ΔT): 10.0 °C
- Conductor length: 15 mm
- Copper resistivity: 1.72e-08 Ω·m

The 'External Layer Tracks' section on the right shows the calculated results:

- Track width (W): 1.36703 mm
- Track thickness (H): 35 μm
- Cross-section area: 0.0478461 mm²
- Resistance: 0.00539229 Ω
- Voltage drop: 0.0161769 V
- Power loss: 0.0485306 W

The 'Internal Layer Tracks' section on the right shows the calculated results:

- Track width (W): 3.55625 mm
- Track thickness (H): 35 μm
- Cross-section area: 0.124469 mm²
- Resistance: 0.00207281 Ω
- Voltage drop: 0.00621843 V
- Power loss: 0.0186553 W

The central text area provides the following information:

If you specify the maximum current, then the track widths will be calculated to suit.

If you specify one of the track widths, the maximum current it can handle will be calculated. The width for the other track to also handle this current will then be calculated.

The controlling value is shown in bold.

The calculations are valid for currents up to 35 A (external) or 17.5 A (internal), temperature rises up to 100 °C, and widths of up to 400 mils (10 mm).

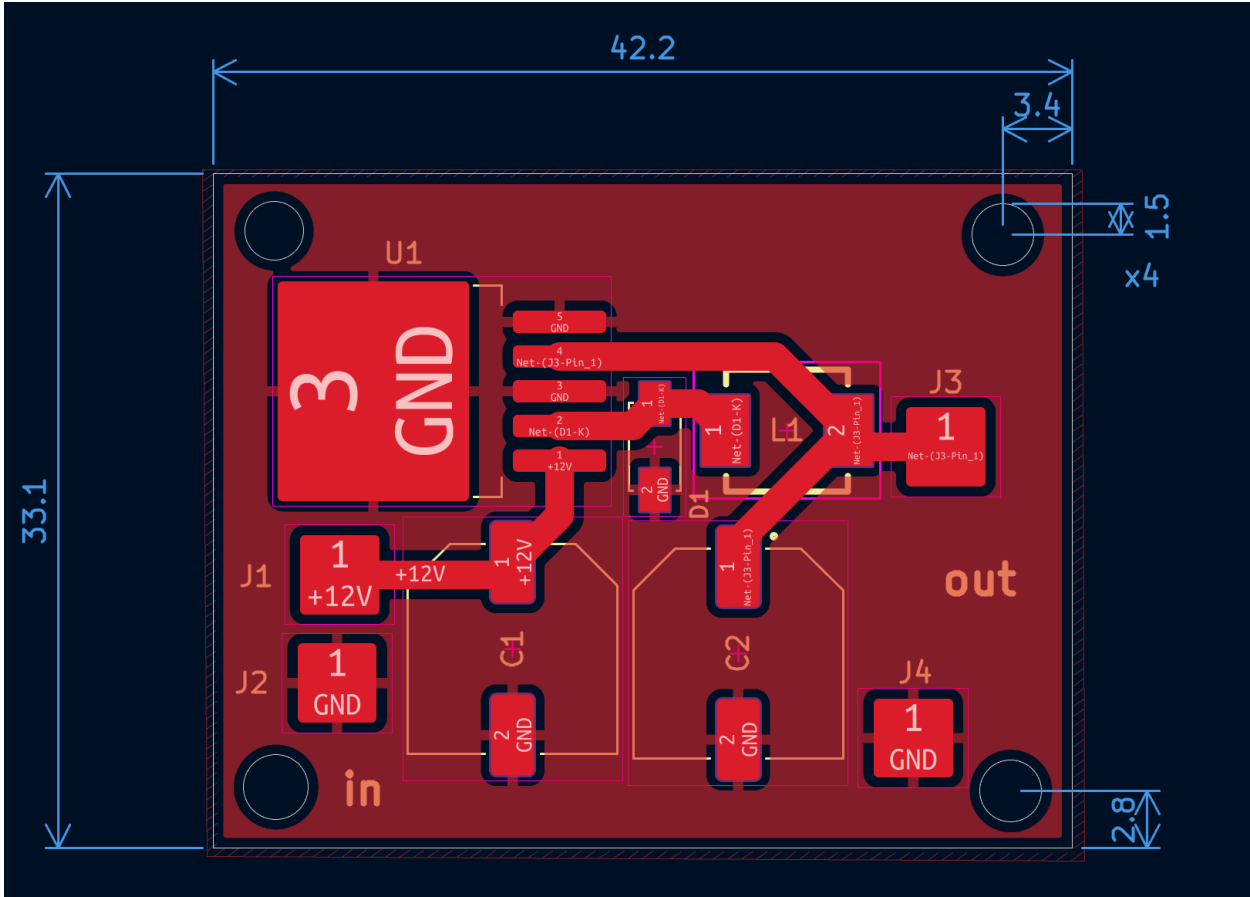
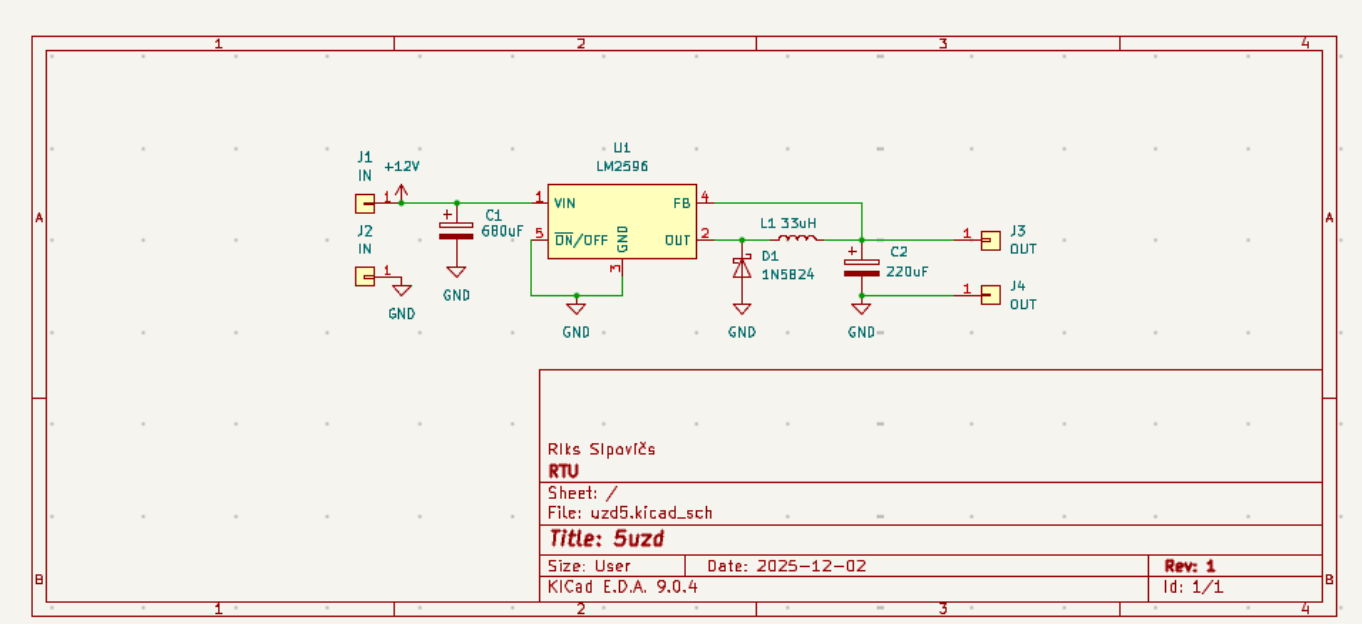
The formula, from IPC 2221, is

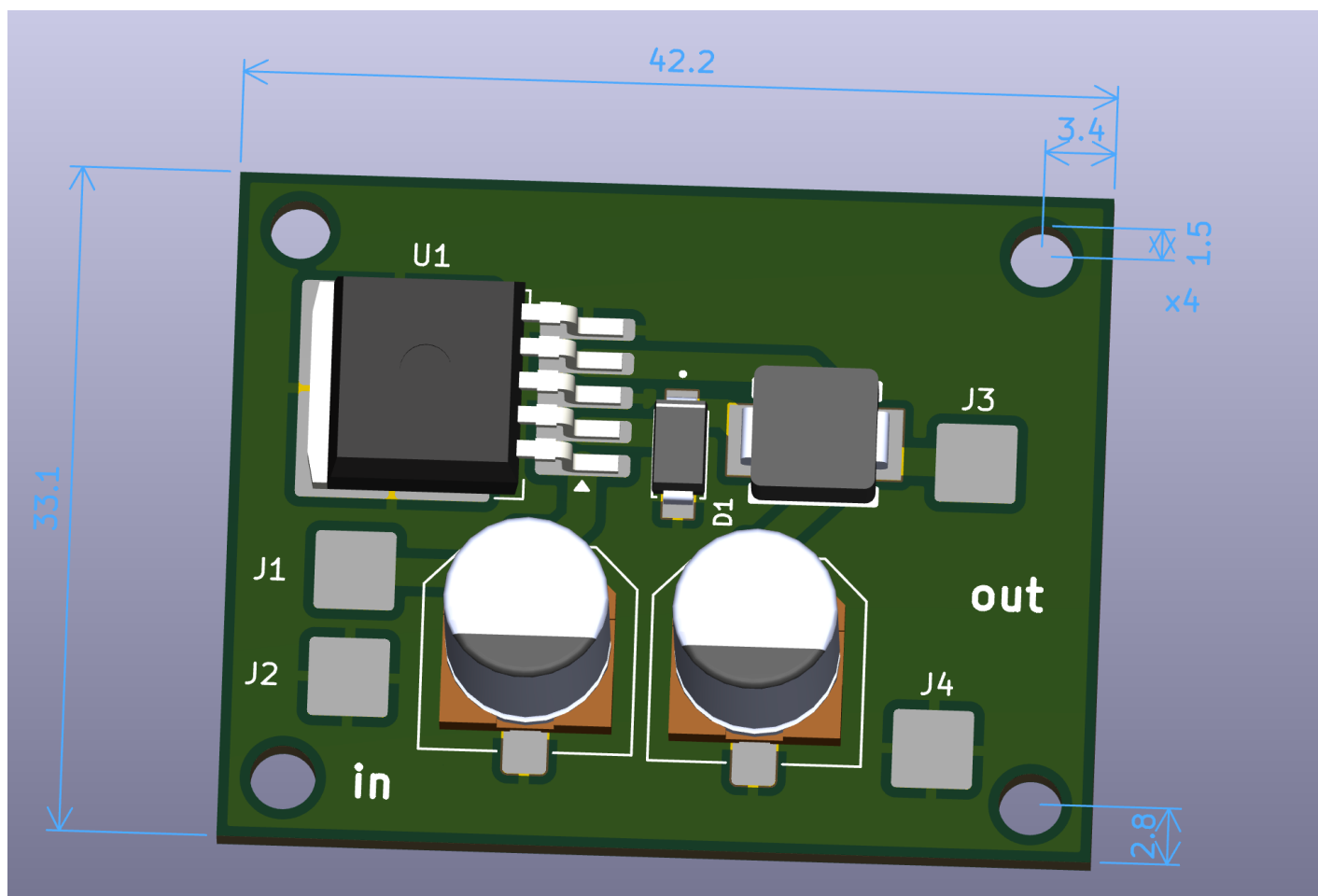
$$I = K \cdot \Delta T^{0.44} \cdot (W \cdot H)^{0.725}$$

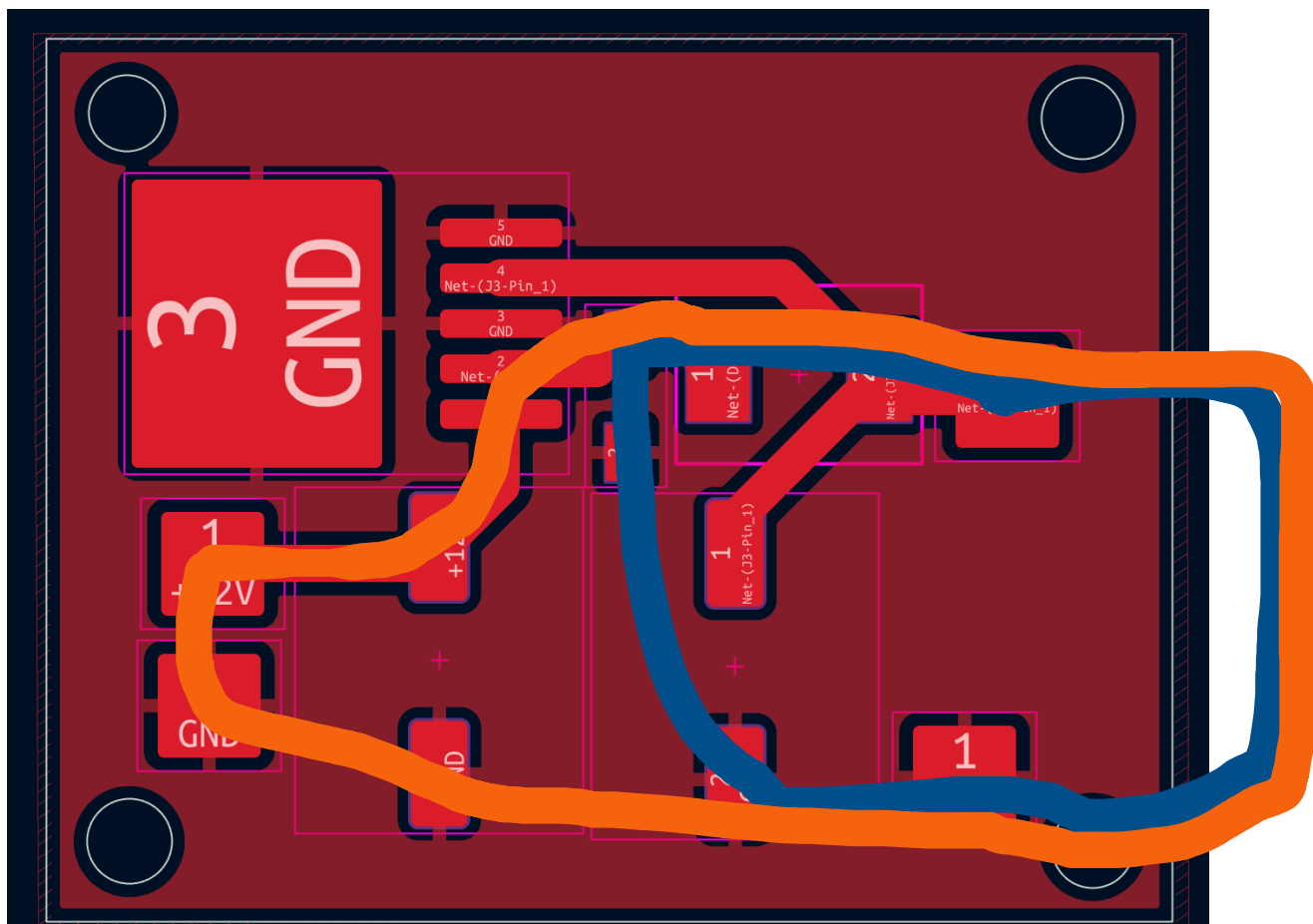
where:

- I is maximum current in A
- ΔT is temperature rise above ambient in °C
- W is width in mils
- H is thickness (height) in mils
- K is 0.024 for internal tracks or 0.048 for external tracks

Paņemu **1,4 mm** celiņa platumu, jo slodze var būt strāva līdz 3A.







Lai samazinātu “loop area” un magnētiskā lauka izraisītās sprieguma svārstības, induktoru, diodi un LM2596 izejas kāju novietoju ļoti tuvu.

Kad slēdzis pārslēdzas, strāvas cilpa strauji mainās, un mazāks cilpas laukums samazina **inducētos traucējumus** un **elektromagnētisko starojumu**, nodrošinot stabilāku pārveidotāja darbību.